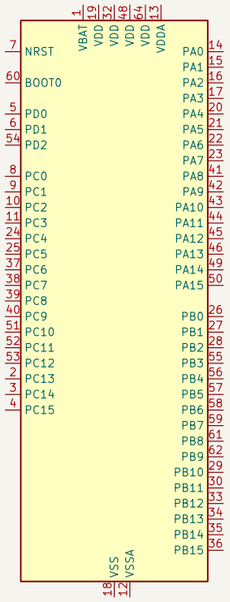
ALPHA 1S V1.1 / 2015-01-04 — ORIGINAL BOARD RECONSTRUCTION

WIP: thirteen photo-traced intercomponent nets; circuit and board recovery incomplete. NOT FOR FABRICATION.

U8: STM32F103RDT6 confirmed by direct donor reading, 2026-10-18. Connections and firmware remain unrecovered.

U8
STM32F103RDT6



Bluetooth region

File: bluetooth-region.kicad_sch

Audio amplifier region

File: audio-amplifier-region.kicad_sch

J2
BAT - mapping pending



J7
Harness - mapping pending



J8
Harness - mapping pending



J9
Harness - mapping pending

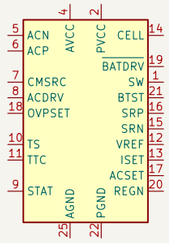


J6
9 contacts, keyed



J6: unpopulated; nine holes, position 10 absent.
Pitch appears near 2 mm; drill size unmeasured.
Intermediate pin sequence inferred; nets unknown.

U3
BQ24133



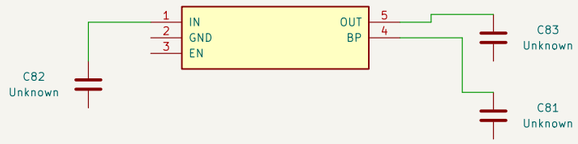
U9
SN74LVC2G125



J10
Harness - mapping pending



U14
SGM2019-3.3



U14: three connections traced from video.
U13 and its capacitors moved to the audio sheet.

Still to identify: U1, other discrete parts, remaining passives, antenna matching and connector variants.

Y1 reads 8.000 MHz; package/pin configuration pending. J3 USB, SPK and LED await mapping.

J2 and J7–J10: pin numbers shown here are temporary schematic numbering.

Confirm physical pin-1 conventions before assigning footprints or nets.

U3: BQ24133: TI Rev. D pin map, including AGND pad 25.
U9/U13/U14: video markings and manufacturer pin maps verified.
Cell setting, supply rails and remaining donor wiring untraced.

No invented wires, no no-connect flags used to hide unknown wiring. ERC warnings are expected at this stage.

Q4
CSD17313Q2



Input switch region

File: input-switch-region.kicad_sch

U4
MS6324G



C68
Unknown

Q1
IRF9310 (P-channel)



Q3
CSD17313Q2



Q2
IRF9310 (P-channel)



Recovered back resistors

File: recovered-resistors-back.kicad_sch

Recovered front resistors

File: recovered-resistors-front.kicad_sch

U4/Q1/Q2: identities from direct donor readings, 2026-10-18.
Q1 source pin3 to C68 traced from video; other local connections unverified.

INCOMPLETE: no donor nets or physical geometry verified

Sheet: /
File: alpha1s-original.kicad_sch

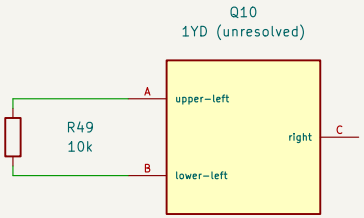
Title: Alpha 1S V1.1 — Original reconstruction

Size: A3	Date: 2026-10-17	**Rev: WIP-01**
KiCad E.D.A. 10.0.6		Id: 1/6

INPUT SWITCH REGION / DONOR CIRCUIT RECOVERY

Q10 A/B/C are photograph-relative terminals, not manufacturer pin numbers.

Source: video frames 138, 140 and 142. Copper seen on the surface; not continuity measured.



R49 connects across Q10 terminals A and B. Terminal C continues to a via; destination unresolved.

Nearby populated resistors: decoded values below; their nets remain to be traced.



No footprint or transistor function is assigned to Q10 until the donor pinout is resolved.

Observed connections only; Q10 functional pinout unresolved

Sheet: /Input switch region/
File: input-switch-region.kicad_sch

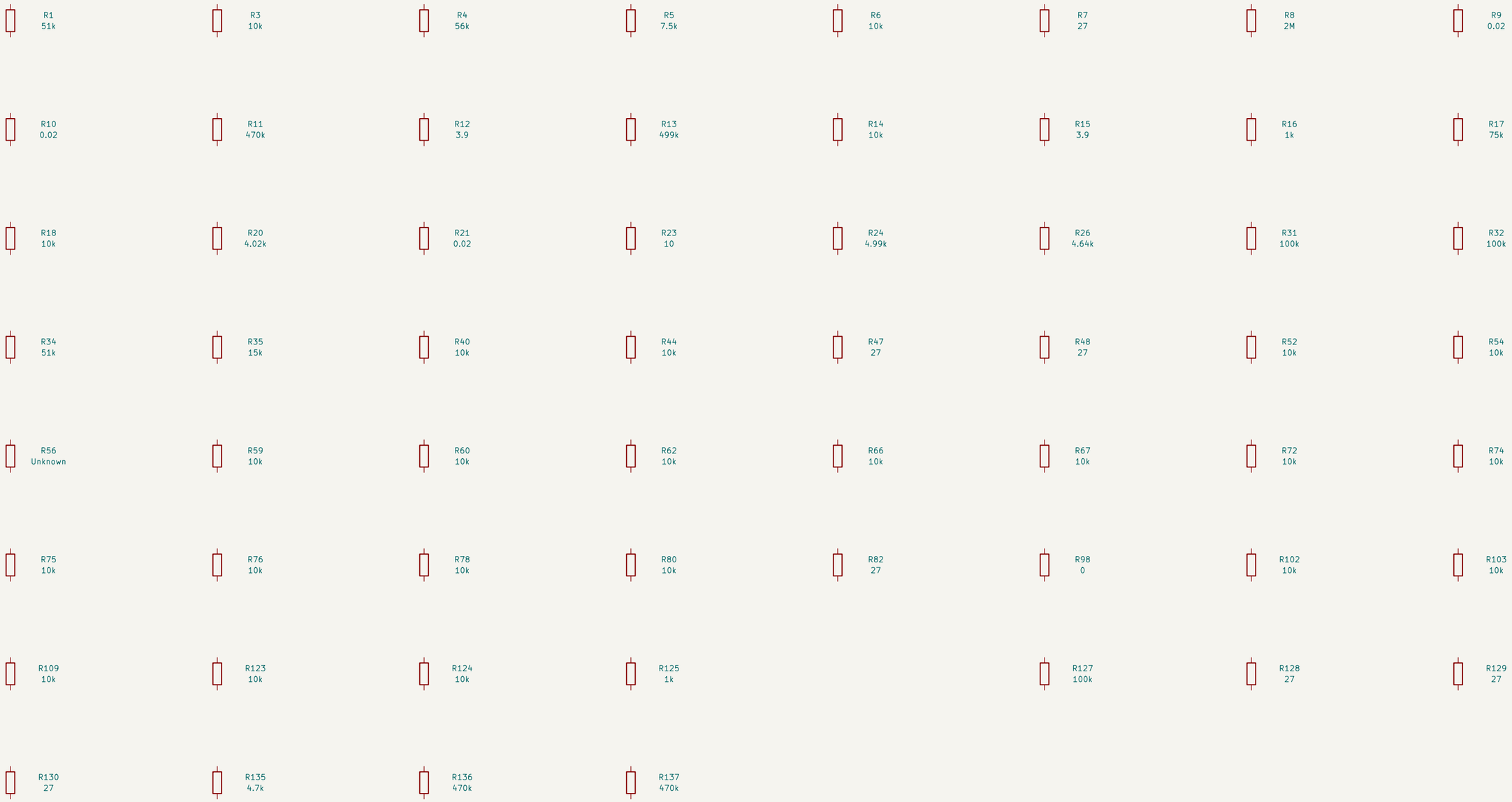
Title: Alpha 1S V1.1 – Input switch region

Size: A4	Date: 2026-10-18	**Rev: Recovery**
KiCad E.D.A. 10.0.6		Id: 2/6

ALPHA 1S V1.1 / BACK RESISTORS — RECOVERED VALUES

Donor markings and references retained. All terminals below await tracing and physical pad mapping.

Unknown values preserve ambiguous readings. Package, tolerance and power rating remain unresolved.



R49/R50/R61/R65 are on the input-switch sheet; R143/R126 are on the amplifier sheet.

R58 is excluded pending resolution of conflicting photograph references.

Values read from donor; connections pending

Sheet: /Recovered back resistors/
File: recovered-resistors-back.kicad_sch

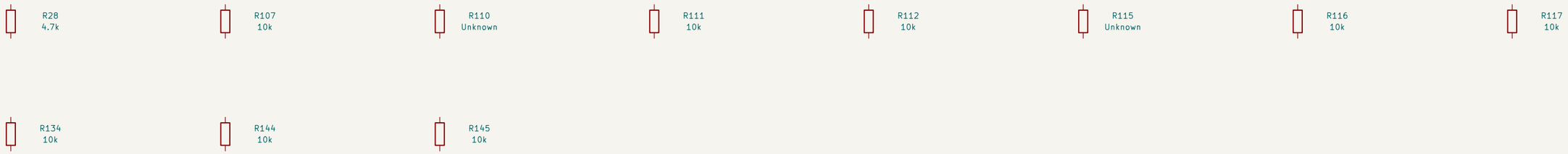
Title: Alpha 1S — recovered back resistors

Size: A3	Date: 2026-10-18	**Rev: Recovery**
KiCad E.D.A. 10.0.6		Id: 3/6

ALPHA 1S V1.1 / FRONT RESISTORS — RECOVERED VALUES

Donor markings and references retained. All terminals below await tracing and physical pad mapping.

Unknown values preserve ambiguous readings. Package, tolerance and power rating remain unresolved.



Parts already present on the input-switch sheet are not repeated here: R49, R50, R61, R65.

R58 is excluded pending resolution of conflicting photograph references.

Values read from donor; connections pending

Sheet: /Recovered front resistors/
File: recovered-resistors-front.kicad_sch

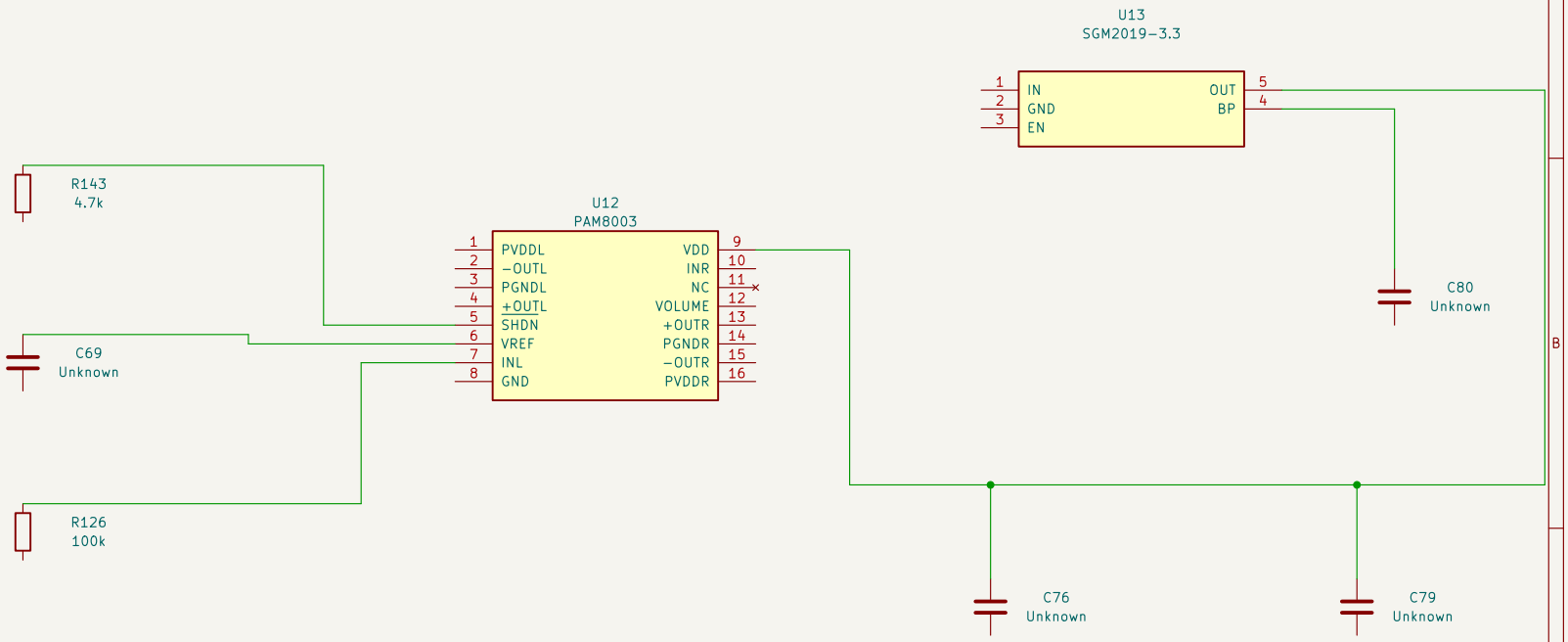
Title: Alpha 1S — recovered front resistors

Size: A3	Date: 2026-10-18	**Rev: Recovery**
KiCad E.D.A. 10.0.6		Id: 4/6

AUDIO AMPLIFIER / OBSERVED DONOR CONNECTIONS

U12 pin 1: package dot at upper-left in video frame007. Pin map: Diodes PAM8003 datasheet.

Observed local traces and U13-to-U12 supply path. Remaining connections unresolved.



Pin1 convention: R143/C69/R126 right pad; C76/C79/C80 left pad in video frame007.

R143 4.7k to shutdown pin5; C69 to reference pin6; R126 100k to left-input pin7.

Observed surface continuity only. Capacitor value and remaining branches are not established.

U13 OUT pin5, C79 pin1, C76 pin1 and U12 VDD pin9 share the observed surface path.

U13 BP pin4 connects to C80 pin1. Capacitor values and opposite terminals remain unknown.

Photo-traced amplifier and regulator connections; remaining wiring unresolved

Sheet: /Audio amplifier region/
File: audio-amplifier-region.kicad_sch

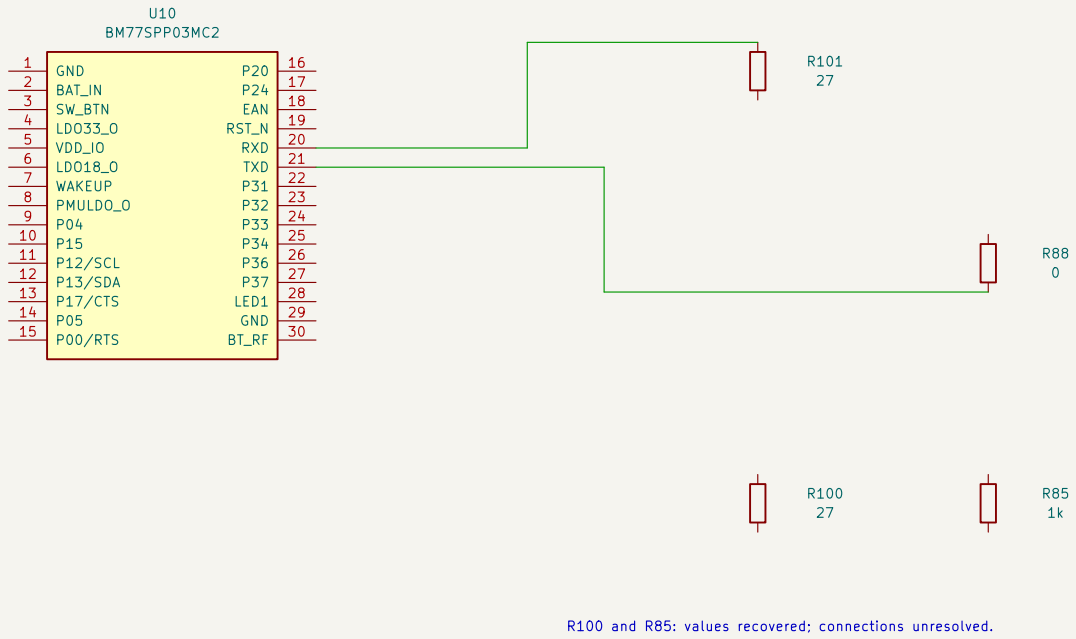
Title: Alpha 1S – Audio amplifier region

Size: A4	Date: 2026-10-18	**Rev: Recovery**
KiCad E.D.A. 10.0.6		Id: 5/6

BLUETOOTH / OBSERVED DONOR CONNECTIONS

U10 pin map: BM77SPP03MC2 datasheet, June 2015, Figure 1–3 and Table 1–1.

Photo–traced local connections only. No MCU destination, supply or ground net inferred.



R100 and R85: values recovered; connections unresolved.

Passive pin1 = left terminal, pin2 = right terminal with donor BM77 label upright.

RXD pin20 to R101 left; TXD pin21 to R88 right. Opposite terminals unresolved.

P12/SCL, P13/SDA and P36 are marked Do Not Connect in the module datasheet.

Two visible local traces; UART destinations and other wiring unresolved

Sheet: /Bluetooth region/
File: bluetooth–region.kicad_sch

Title: Alpha 1S – Bluetooth region

Size: A4	Date: 2026–10–20	**Rev: Recovery**
KiCad E.D.A. 10.0.6		Id: 6/6